AMINE QUIROGA BRITOS

Software Engineer | Rust • Systems Programming
San Luis, Argentina

PROFESSIONAL SUMMARY

Computer Engineer (95% completed, Thesis in progress) with 2 years of professional experience developing production software in Rust. Experienced in designing high-performance backend services, asynchronous systems, and scalable APIs, supported by a strong academic background in computer architecture, embedded systems, FPGA development, and computer graphics. Passionate about systems programming, performance optimization, and low-level software engineering.

TECHNICAL SKILLS

- **Languages:** Rust, C++, C, TypeScript, JavaScript, SQL, Go, Python, VHDL, Verilog
- **Backend & Systems:** Rust (Tokio, Axum, sqlx), Node.js, Express, REST APIs, OpenAPI/Swagger, PostgreSQL, MySQL, Prisma ORM
- **Frontend:** Svelte, SvelteKit, HTML, CSS, Tailwind CSS, Skeleton UI
- **Graphics & Real-Time:** WebGL2, GLSL, Three.js, Unity, Path Tracing, BVH
- **Embedded & Hardware:** ARM Cortex-M4, FPGA, AMD-Xilinx Vitis, Vivado, HLS, Digital Logic Design
- **Cloud & DevOps:** AWS S3, CloudWatch, Git, GitHub Actions, Linux
- **Tools:** Postman, Jira, Figma, LaTeX

PROFESSIONAL EXPERIENCE

Fullstack Software Engineer (Rust-focused)
DC Connected Car GmbH
02/2024 – 01/2026

- Developed and maintained production backend services in **Rust (Axum and Tokio)**, implementing asynchronous REST APIs and business logic for scalable web applications.
- Designed and optimized **PostgreSQL** data models and queries using **sqlx**, leveraging compile-time SQL verification and controlled database migrations.
- Improved existing services by refactoring codebases for maintainability and maintaining **OpenAPI/Swagger** documentation to support cross-team development.
- Integrated **AWS S3** for file storage and retrieval, implementing secure and efficient data handling workflows.
- Monitored production environments using **AWS CloudWatch** and **ECS**, investigating logs and debugging application issues in deployed services.
- Built and maintained frontend features using **SvelteKit**, **TypeScript**, and **Tailwind CSS**, contributing reusable UI components and improving overall user experience.
- Integrated third-party services including the **OpenAI API** and **ElevenLabs API** to deliver AI-powered product features.
- Contributed to **Git-based development workflows**, participating in code reviews, CI/CD pipelines with **GitHub Actions**, Agile ceremonies, and unit/integration testing.

Software Developer
Municipal Administrative Court of Accounts of San Luis
10/2022 – 09/2023

- Developed an automated bank reconciliation system using **Node.js**, **Express**, **MySQL**, and **Prisma ORM**, significantly reducing manual accounting work.
- Implemented PDF parsing and data extraction pipelines, transforming unstructured financial documents into structured transaction records.
- Applied regular expressions and custom parsing algorithms to reconstruct banking statements and automate reconciliation workflows.
- Designed and maintained relational database schemas and backend services for reliable storage and retrieval of financial data.
- Worked closely with accounting staff to gather requirements, validate reconciliation results, and refine business logic.
- Built web interfaces using **HTML**, **CSS**, and **Bootstrap** to support reconciliation and reporting tasks.
- Provided workstation, operating system, and local network support for the organization's IT infrastructure.

SELECTED PROJECTS

Real-Time Path Tracing Renderer (Graduation Thesis)
National University of San Luis
2024 – Present

- Implemented a physically based path tracer in WebGL2 and GLSL from scratch.
- Designed GPU acceleration structures (BVH) and optimized traversal algorithms for real-time rendering.
- Implemented Monte Carlo rendering techniques including importance sampling, MIS, Next Event Estimation and Russian Roulette.
- Focused on GPU performance optimization and progressive rendering for interactive applications.

RACube (Augmented Reality Educational Application)
National University of San Luis
2023 – 2024

- Awarded a competitive university research scholarship.
- Designed and developed a complete augmented reality educational application using Unity and Vuforia.
- Responsible for the entire software lifecycle, from UX design to deployment.

EDUCATION

B.S. in Computer Engineering
National University of San Luis
2019 – Present

- **Status:** All coursework completed. Currently completing graduation thesis.
- **Relevant Coursework:** Computer Architecture, Operating Systems, Computer Graphics, Embedded Systems, ARM Cortex-M4 Microcontrollers, Digital Logic Design, FPGA Development, Concurrent Programming.

CONTINUING EDUCATION & CERTIFICATIONS

FPGA & Hardware

- FPGA Vision Open Online Course
- XI Southern Programmable Logic Conference
    - SoC-FPGA Design Flow
    - FPGA Accelerated Cloud Computing with AMD-Xilinx Vitis
    - Versal ACAP AI Engine Programming

Software Engineering

- REST APIs with Go
- Scientific Writing with LaTeX

Design

- UX/UI Design Intensive Program

LANGUAGES

- **Spanish:** Native speaker
- **English:** Fluent (Professional and Technical Proficiency)
- **Japanese:** Basic